Structured Reasoning Frameworks as Cost-Efficiency Multipliers

An empirical evaluation of DODAR scaffolding across eight language models, five prompting conditions, and ten decision scenarios with dual-evaluator scoring

Adam Field

March 2026

GPT-4.1 Mini + DODAR pipeline scores 4.80/5.0 (104% of Opus zero-shot) at 89% lower cost per query

Abstract

This paper evaluates whether structured decision-making frameworks can close the quality gap between small language models and frontier models on complex reasoning tasks. DODAR (Diagnose, Options, Decide, Action, Review) is a five-phase framework originating from aviation Crew Resource Management. We benchmark eight models from four capability tiers across five prompting conditions and six quality dimensions, using ten scenarios spanning business, medical, legal, technical, and personal domains. All responses are scored independently by two evaluators (Claude Opus 4.6 and GPT-5.4) to control for evaluator-model bias, with scores averaged across both. Three key findings emerge. First, DODAR scaffolding benefit is inversely proportional to model capability: small models gain up to +1.45 points on a 5-point scale (Cohen's d = 2.78) while frontier models show marginal improvement. Second, a five-phase pipeline implementation where each phase runs as a separate model call with accumulated context consistently outperforms the single-prompt approach. GPT-4.1 Mini with the DODAR pipeline scores 4.80, reaching 104% of Claude Opus 4.6 zero-shot quality (4.62) at 89% lower real API cost. Third, chain-of-thought prompting underperforms zero-shot for six of eight models under dual scoring, challenging assumptions about generic reasoning elicitation. These findings are subject to significant methodological caveats including ceiling effects, inconsistent sample sizes, and untested prompt sensitivity, all of which are discussed in detail.

---

1. Introduction

The default approach to AI-assisted decision-making is to use the most capable model available and rely on its native reasoning. Frontier models like Claude Opus 4.6 and GPT-5.4 achieve strong scores on structured decision tasks with zero-shot prompting. But frontier API pricing is dramatically higher than small model equivalents. In this evaluation, a single Opus zero-shot query costs $0.1424 on average, compared to $0.000334 for GPT-4.1 Nano. For high-volume production workloads, this premium compounds quickly.

This paper tests a different hypothesis: rather than paying for a more capable model, can you achieve equivalent results by giving a cheaper model better structure? We use DODAR as the test framework and evaluate two implementations: a single-prompt approach where all five phases are specified in one instruction, and a pipeline approach where each phase runs as a separate model call with the output of prior phases fed forward as context.

The evaluation uses dual-evaluator scoring throughout. Every response is scored independently by Claude Opus 4.6 and GPT-5.4, with scores averaged. This design partially addresses the evaluator-model overlap problem inherent in using a language model to evaluate language model outputs. We report inter-rater reliability statistics alongside all findings.

2. The DODAR Framework

DODAR is a structured decision-making framework developed for aviation Crew Resource Management (CRM). It was designed for cockpit environments where premature action under uncertainty leads to fatal outcomes. The framework enforces five sequential phases, each acting as a cognitive gate that must be cleared before proceeding.

Diagnose. Hold the diagnosis open. List competing hypotheses rather than anchoring on the most obvious explanation. Surface assumptions. Identify what information is missing. Challenge the first answer that presents itself.

Options. Generate at least four genuinely distinct alternatives. Name the core tensions and trade-offs explicitly. Quantify opportunity costs where possible. Include unconventional or counterintuitive options.

Decide. Commit to a recommendation with transparent reasoning. Justify the choice against specific alternatives. State binding constraints, confidence levels, and the conditions under which the decision should be revisited.

Action. Translate the decision into concrete, sequenced steps. Identify dependencies and blockers. Distinguish reversible from irreversible actions. Specify timelines and resource requirements.

Review. Identify at least three failure modes with detection methods and contingency plans. Validate assumptions made during earlier phases. Specify conditions for abandoning the plan entirely.

The hypothesis underlying this study is that language models fail at complex decision tasks in the same ways humans fail under pressure: through premature closure, option narrowing, and skipping critical reasoning steps. DODAR compensates for these failures by externalising the decision process into an explicit scaffold.

3. Methodology

3.1 Models Under Test

Eight models were evaluated, spanning four capability tiers across Anthropic and OpenAI model families. This selection covers the practical range of API-available models at the time of testing (March 2026) but does not include models from Google, Meta, or Mistral, limiting generalisability to the two model families tested.

Model	Tier	Input $/1M tokens	Output $/1M tokens
GPT-4.1 Nano	Nano	$0.10	$0.40
GPT-4o Mini	Small	$0.15	$0.60
GPT-4.1 Mini	Small	$0.40	$1.60
Haiku 4.5	Small	$0.80	$4.00
GPT-4o	Mid	$2.50	$10.00
Sonnet 4.5	Mid-Frontier	$3.00	$15.00
GPT-5.4	Frontier	$10.00	$30.00
Opus 4.6	Frontier	$15.00	$75.00

Table 1. Models and API pricing from public rate cards, March 2026.

3.2 Test Scenarios

The scenario bank defines twenty scenarios across two categories, but only ten were fully evaluated across all models and conditions. The ten core scenarios (AMB-01 to AMB-05 and TRD-01 to TRD-05) form the basis of all reported results. An additional seven scenarios received partial runs with only one or two models and were not scored. Three scenarios were never run. All aggregate statistics in this paper are derived from the ten fully evaluated scenarios only. This limited scenario count is a significant constraint on generalisability.

The first five scenarios (AMB-01 to AMB-05) test ambiguous diagnosis: situations where multiple plausible root causes cluster temporally and the correct response is to resist anchoring on a single explanation. These span business (SaaS churn analysis), medical (chest pain differential), legal (clinical negligence causation), technology (production errors), and personal (sleep quality) domains. Difficulty ranges from easy to hard.

The second five (TRD-01 to TRD-05) test competing trade-offs: decisions where multiple valid paths exist and the correct response is to surface genuine tensions rather than default to one option. These cover startup resource allocation, career transitions, public policy (green belt housing), investment selection, and build-vs-buy decisions. Again, difficulty ranges from easy to hard.

Each scenario includes expected pitfalls (common reasoning errors to identify), gold standard elements (what a quality response should contain), and DODAR-specific discriminators (framework-aligned behaviours that distinguish strong from weak responses). Scenarios were authored for this study and have not been independently validated. With only ten scenarios, there is limited coverage of any single domain and no basis for analysing domain-level effects.

3.3 Experimental Conditions

Each model was evaluated under five prompting conditions.

Zero-shot. The baseline. The scenario is presented with a brief instruction to analyse it. No reasoning strategy is specified.

Chain-of-thought (CoT). The model is instructed to think step by step before reaching conclusions.

DODAR (single prompt). The full DODAR framework is provided as a single system instruction. The model is told to work through all five phases sequentially in one response. The prompt specifies detailed requirements for each phase, including minimum hypothesis counts, option counts, and failure mode counts.

DODAR (pipeline). Each DODAR phase runs as a separate model call. Each phase has a specialised system prompt with a role-specific persona: the Diagnose phase uses a "diagnostic reasoning specialist", the Options phase uses a "strategic options analyst", the Decide phase uses a "decision architect", the Action phase uses an "implementation planning specialist", and the Review phase uses a "critical review analyst". The output of each phase is appended to an accumulated context window that feeds into the next phase. Gate checks between phases validate that prior phases met their minimum requirements before proceeding. This architecture means the pipeline consumes substantially more tokens than a single call, as input context grows with each phase.

Length-matched. A control condition where the model is instructed to produce a response of similar length to typical DODAR outputs, using filler instructions about thoroughness, but without the framework structure. This isolates whether DODAR benefits come from the structure itself or simply from eliciting longer responses.

3.4 Scoring Dimensions

Responses were scored on a 1 to 5 Likert scale across six dimensions. Diagnosis Quality measures the systematic enumeration of root causes and acknowledged uncertainty, where 1 indicates surface symptoms only and 5 indicates systematic hypothesis generation with explicit uncertainty. Option Breadth measures the range and distinctness of alternatives generated, where 1 indicates a single option presented and 5 indicates five or more distinct alternatives with explicit opportunity cost articulation. Decision Justification measures the transparency of reasoning behind the recommendation, where 1 indicates unstated preferences and 5 indicates explicit trade-off weighing. Action Specificity measures the concreteness and sequencing of implementation steps, where 1 indicates vague steps and 5 indicates dependency maps with blockers and reversibility assessment. Review and Self-Correction measures the identification of failure modes and contingencies, where 1 indicates no reflection and 5 indicates three or more failure modes with correction triggers. Overall Trustworthiness measures whether an expert would endorse the reasoning quality, where 1 indicates an expert would reject the output and 5 indicates rigorous reasoning an expert would endorse.

The rubric explicitly instructs: "Score the REASONING QUALITY, not whether you agree with the conclusion." A score of 3 is calibrated as "good but with gaps", not average. Responses were presented to evaluators without condition labels to prevent scoring bias.

3.5 Dual-Evaluator Design

A known weakness in LLM evaluation is evaluator-model overlap: using Claude to score Claude's own outputs may introduce systematic bias. To control for this, every response was scored independently by two evaluators: Claude Opus 4.6 and GPT-5.4. Final scores are the average of both evaluators' ratings.

Across 1200 paired comparisons, the two evaluators show 70.5% exact agreement and 99.7% agreement within one point. The mean absolute difference is 0.30 points. Claude scores marginally higher than GPT-5.4 on average (signed difference: +0.08). Agreement is highest on Review and Self-Correction (80% exact) and lowest on Overall Trustworthiness (58% exact). The relatively low agreement on Overall Trustworthiness suggests evaluators may weight different aspects of reasoning quality when making holistic judgements.

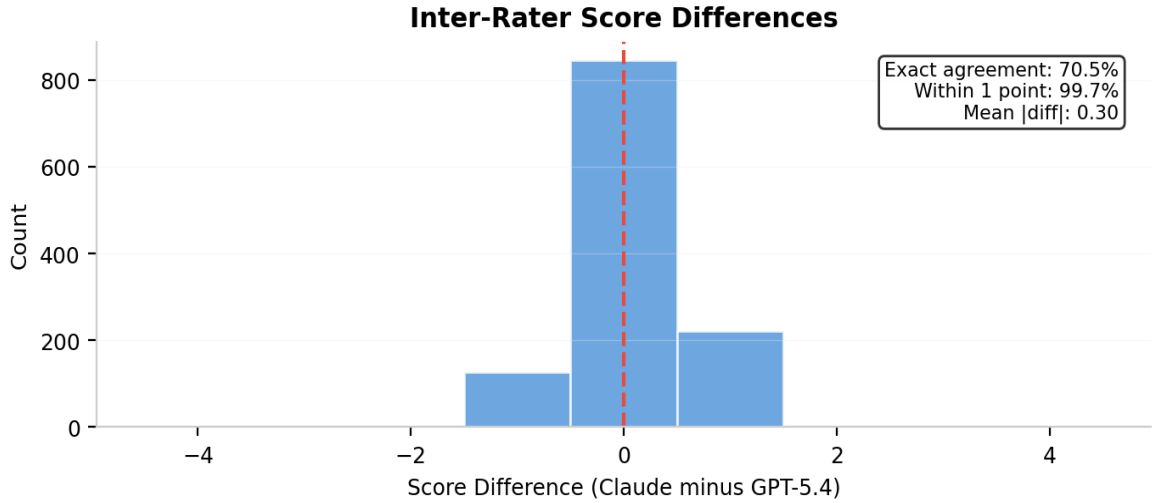

Figure 1. Distribution of inter-rater score differences (Claude minus GPT-5.4). Positive values indicate Claude scored higher.

This design does not fully solve the overlap problem. Claude is still one of two evaluators scoring Anthropic model outputs, and GPT-5.4 is one of two evaluators scoring OpenAI model outputs. A cleaner design would use a third-party evaluator or human judges. The dual-evaluator approach is a partial mitigation, not a full solution.

3.6 Sample Sizes and Data Completeness

The full dataset contains 449 individual response items across all scenarios. Of these, 400 items from the 10 core scenarios were scored. Each model-condition combination has 10 response items (one per scenario). Each item is scored independently by both evaluators, yielding n=20 individual scores per model-condition-dimension cell. The DODAR single-prompt condition has n=40 because those items were scored in two separate evaluation runs per evaluator (an initial run and a follow-up comparing DODAR vs pipeline). Some CoT items also have n=40 for the same reason.

This means n=20 reflects 10 underlying response items scored twice, not 20 independent responses. The effective sample size for independence is 10. This is a small base for drawing robust conclusions. The DODAR single-prompt condition's n=40 does not represent more response diversity, only more evaluation passes over the same 10 responses. Standard deviations and means should be interpreted with this in mind.

49 items from partially-run scenarios (AMB-06 to AMB-10, TRD-06, TRD-07) remain unscored and are excluded from all reported results. Three further scenarios (TRD-08 to TRD-10) were never run.

These gaps are a limitation: the evaluated scenario set covers only five of each category, with limited domain diversity.

4. Results

4.1 Overview: All Conditions

Figure 2 shows mean quality scores across all models and conditions. The grand mean by condition is: zero-shot 3.69, chain-of-thought 3.62, DODAR single-prompt 3.95, length-matched 3.96, and DODAR pipeline 4.46. The pipeline condition leads by a substantial margin. Two patterns are immediately visible: the gap between small and frontier models under zero-shot conditions is large (ranging from 2.89 to 4.62), and DODAR scaffolding, particularly the pipeline variant, substantially narrows this gap.

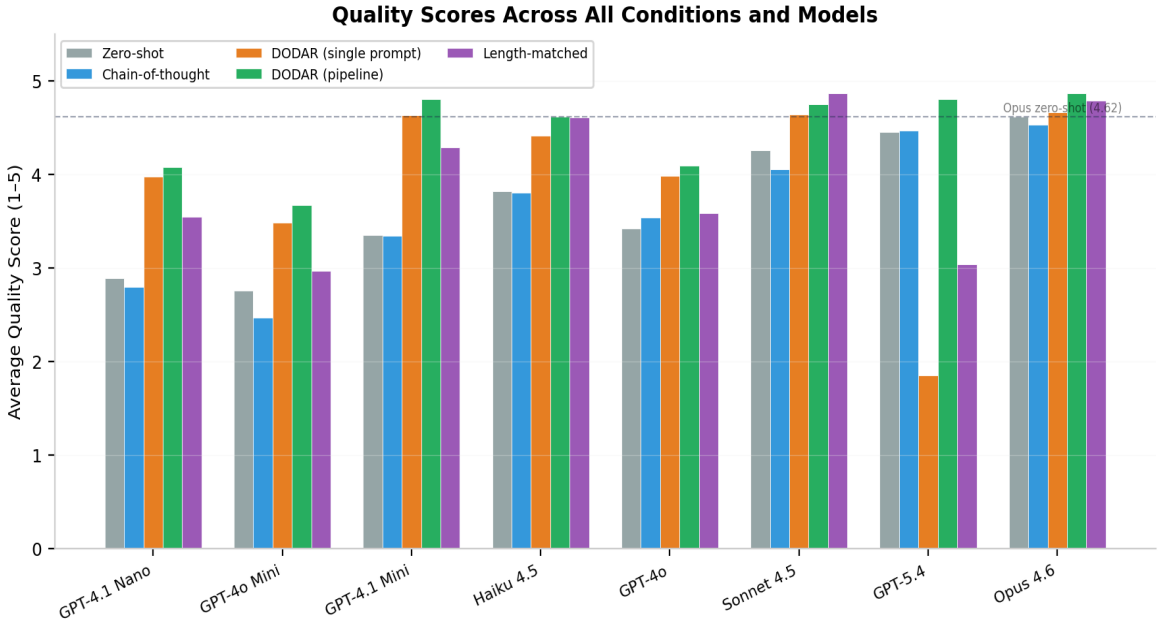

Figure 2. Mean quality scores (average across six dimensions) for all models and conditions.

Model	Zero-shot	CoT	DODAR	Pipeline	Length-m.
GPT-4.1 Nano	2.89	2.79	3.97	4.08	3.54
GPT-4o Mini	2.76	2.47	3.48	3.67	2.97
GPT-4.1 Mini	3.35	3.34	4.63	4.80	4.28
Haiku 4.5	3.82	3.80	4.41	4.62	4.61
GPT-4o	3.42	3.53	3.98	4.09	3.58
Sonnet 4.5	4.26	4.05	4.64	4.75	4.87
GPT-5.4	4.45	4.47	1.85	4.80	3.03
Opus 4.6	4.62	4.53	4.66	4.87	4.78

Table 2. Mean quality scores by model and condition (averaged across six dimensions and both evaluators).

4.2 The Scaffolding Effect

Figure 3 compares the DODAR lift (improvement over zero-shot) for both the single-prompt and pipeline implementations. The inverse relationship between model capability and scaffolding benefit is visible in both, but the pipeline consistently outperforms the single prompt.

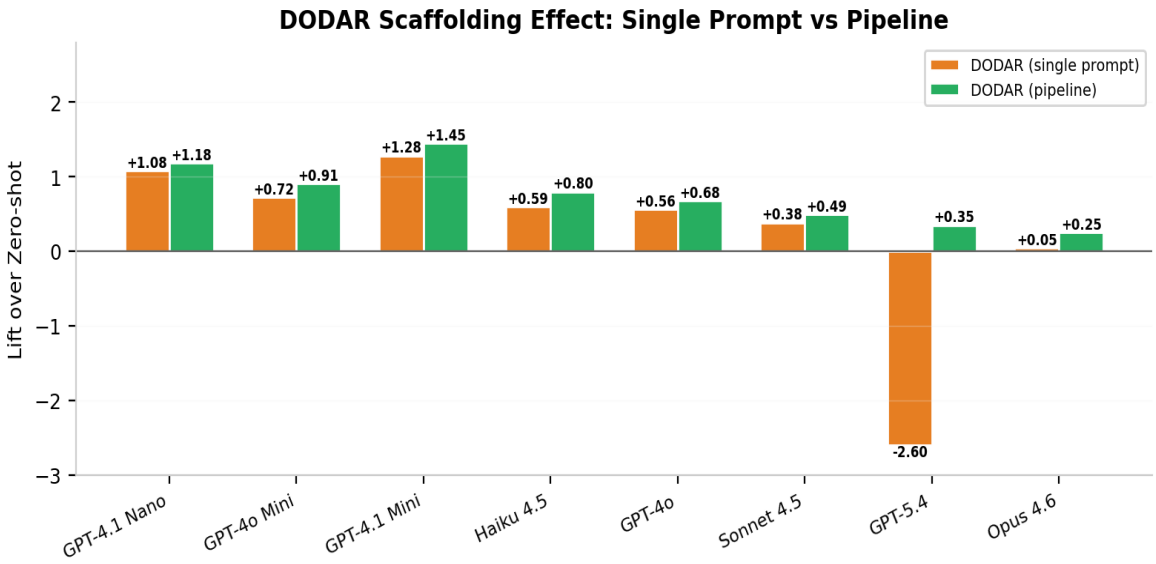

Figure 3. DODAR lift over zero-shot baseline. Orange = single prompt, green = pipeline.

The pipeline produces the largest gains for small models: GPT-4.1 Mini +1.45 (Cohen's d = 2.77), GPT-4.1 Nano +1.18 (d = 2.13), and Haiku 4.5 +0.80 (d = 0.94). These are large to very large effect sizes by conventional thresholds (d > 0.8). Frontier models show smaller gains: Opus 4.6 +0.25 (d = 0.15), a negligible effect.

The effect sizes reported here use the DODAR single-prompt condition as the treatment, compared against each baseline condition. The pipeline effect sizes would likely be larger, but the exported effect size data does not include pipeline-specific calculations. This is a gap in the analysis.

4.3 Per-Dimension Analysis: Pipeline

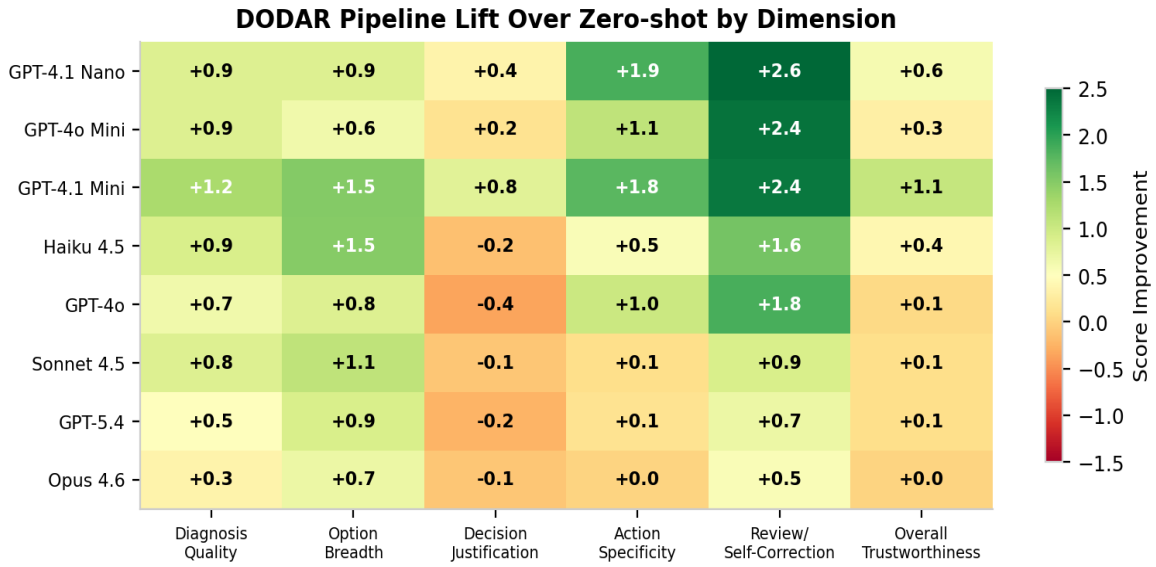

Figure 4. Pipeline lift over zero-shot by model and dimension. Green = improvement, red = degradation.

Averaged across all models with pipeline data, the largest improvements are in Review / Self-Correction (+1.62), Option Breadth (+1.01), and Action Specificity (+0.81). Review and Self-Correction consistently shows the strongest response to pipeline scaffolding because small models simply do not self-critique without an explicit review phase. The pipeline forces this step.

Action Specificity also responds well to the pipeline. The dedicated Action phase with its "implementation planning specialist" persona produces more concrete and sequenced outputs than asking for actions as part of a monolithic response.

However, Decision Justification shows inconsistent behaviour. Several models show pipeline degradation on this dimension: the pipeline may fragment decision reasoning across the Decide and Review phases, diluting the justification quality compared to a single coherent response. This is a design limitation of the pipeline architecture that warrants attention.

4.4 The Chain-of-Thought Paradox

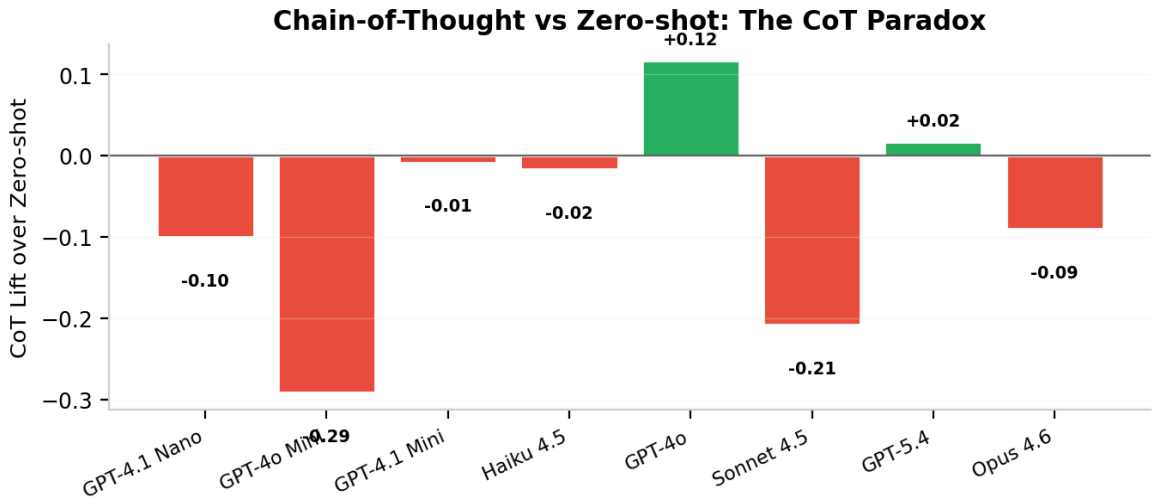

Figure 5. Chain-of-thought lift over zero-shot. Red bars indicate CoT underperformed.

Under dual scoring, chain-of-thought prompting underperforms zero-shot for 6 of 8 models. This is unexpected. The standard assumption is that CoT either helps or is neutral. The most likely explanation is that dual scoring, by introducing a second evaluator with different preferences, deflates scores for verbose outputs more than for concise ones. CoT tends to produce longer, more discursive responses that may be penalised by one evaluator while rewarded by the other. It is also possible that CoT genuinely underperforms zero-shot on this task type, where the problem is not computational complexity but decision structuring. Either way, this finding warrants replication before strong conclusions are drawn.

4.5 The GPT-5.4 Collapse

GPT-5.4 produced the most dramatic anomaly in the dataset. Its zero-shot score of 4.45 is competitive with other frontier models. Under the single DODAR prompt, it collapses to 1.85, a 59% degradation. The length-matched condition also underperforms at 3.03, suggesting this is not specific to the DODAR framework but reflects a broader sensitivity to verbose or heavily structured prompts. The pipeline restores performance to 4.80, exceeding the zero-shot baseline.

GPT-5.4 also shows the highest variance in the dataset: standard deviations reach 1.83 under the length-matched condition and 1.62 under single DODAR. By comparison, most other model-condition cells have standard deviations below 0.70. This extreme variance suggests the model's behaviour under these conditions is not merely degraded but unstable, producing a bimodal distribution of strong and weak responses rather than consistently poor ones.

With only 10 underlying response items per condition (scored by two evaluators to produce n=20 scores), it is difficult to determine whether this collapse is a stable model property or a sampling artefact. Replication with more scenarios and prompt variants is essential before drawing strong conclusions.

4.6 The Sonnet Anomaly

Claude Sonnet 4.5 presents a unique pattern where the length-matched control (4.87) outperforms the pipeline (4.75). This is the only model where the control condition beats the structured intervention. The most parsimonious explanation is that Sonnet operates near its performance ceiling on these tasks, and the marginal benefit of DODAR structure is outweighed by the cost of framework compliance overhead. The length-matched condition gives the model room to reason freely at greater length, which for a near-frontier model may be more productive than imposed structure.

4.7 Ceiling Effects

The 5-point scale shows compression at the top. Across all 240 model-condition-dimension cells, 24 (10.0%) score 4.90 or above, and 13 (5.4%) hit the maximum of 5.00. The pipeline condition accounts for a disproportionate share: 14 of 48 pipeline cells (29.2%) score 4.90 or above.

This ceiling effect has two implications. First, it means the pipeline's true benefit may be underestimated for models that are already hitting the scale maximum. If a model scores 5.0 under both pipeline and some other condition, we cannot tell whether the pipeline would have produced better reasoning on a more discriminating scale. Second, it makes it harder to distinguish between pipeline and frontier zero-shot performance, since both cluster near the ceiling. A 7- or 10-point scale, or a rubric with more granular anchors at the top end, would improve discrimination in future work.

4.8 Variance and Reliability

The pipeline condition shows substantially lower variance than other conditions. The mean standard deviation across all pipeline cells is 0.36, compared to 0.67 for zero-shot, 0.66 for CoT, and 0.67 for length-matched. This suggests the pipeline not only raises average performance but makes outputs more consistent, which is a desirable property for production deployment.

The exception is GPT-5.4, where variance under certain conditions is extreme (std up to 1.83). Excluding GPT-5.4, the highest standard deviation in the dataset is 1.12, demonstrating that this model is an outlier in terms of response stability.

4.9 Cost and Latency

Model + Pipeline	Quality	% Opus ZS	$/Query	vs Opus ZS	Latency
GPT-4.1 Nano	4.08	88%	$0.0024	98% cheaper	39s
GPT-4.1 Mini	4.80	104%	$0.0155	89% cheaper	102s
Haiku 4.5	4.62	100%	$0.1030	28% cheaper	227s
GPT-4o	4.09	89%	$0.0504	65% cheaper	48s
Opus 4.6 (zero-shot)	4.62	100%	$0.1424	baseline	47s

Table 3. Real cost and latency from API calls during benchmarking, not estimates.

Cost data in Table 3 is derived from actual API charges recorded during the benchmark runs, not from estimates. The pipeline token multiplier varies significantly by model: GPT-4.1 Nano uses 13x the tokens of its zero-shot call, while Haiku 4.5 uses 49x. This variation reflects differences in model verbosity and how much context accumulates across pipeline phases. Anthropic models tend to produce substantially longer outputs per phase, inflating the accumulated context faster.

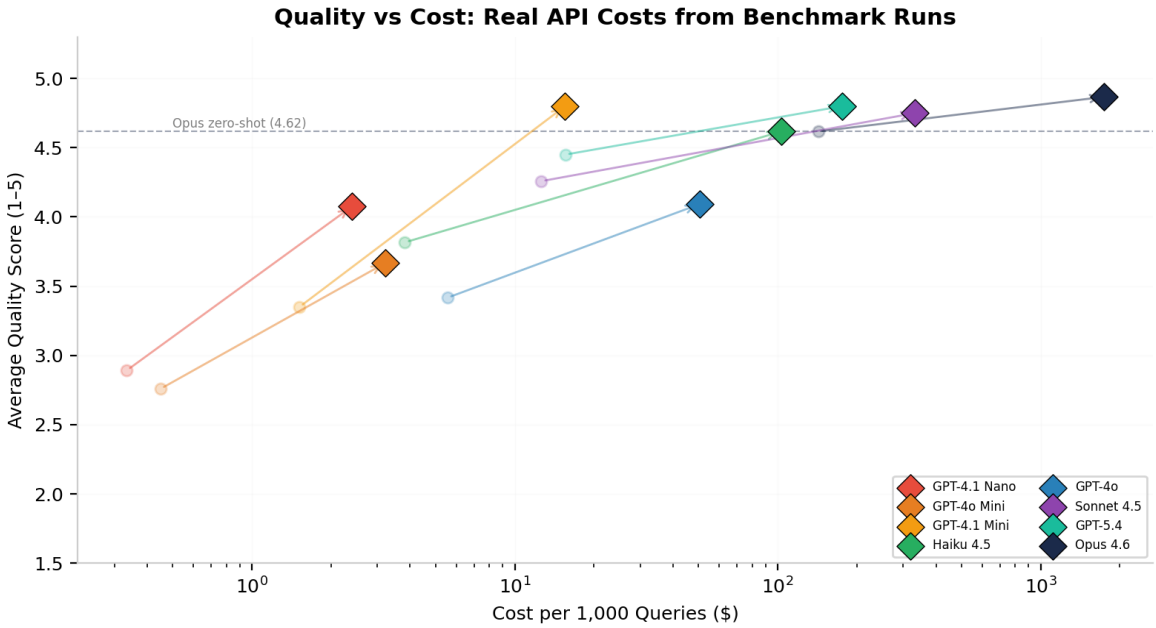

Figure 6. Quality vs real API cost per 1,000 queries. Diamonds = pipeline, faded circles = zero-shot. Arrows show improvement direction. Dashed line = Opus zero-shot baseline.

The cost efficiency headline remains intact despite the higher-than-expected token multipliers. GPT-4.1 Mini pipeline achieves 104% of Opus zero-shot quality at $0.0155 per query, compared to $0.1424 for Opus. At 100,000 queries per month, this translates to approximately $1554 versus $14236, a saving of $12682 per month.

However, latency is a significant practical constraint. The pipeline requires sequential API calls, and the cost of this sequentiality varies by model.

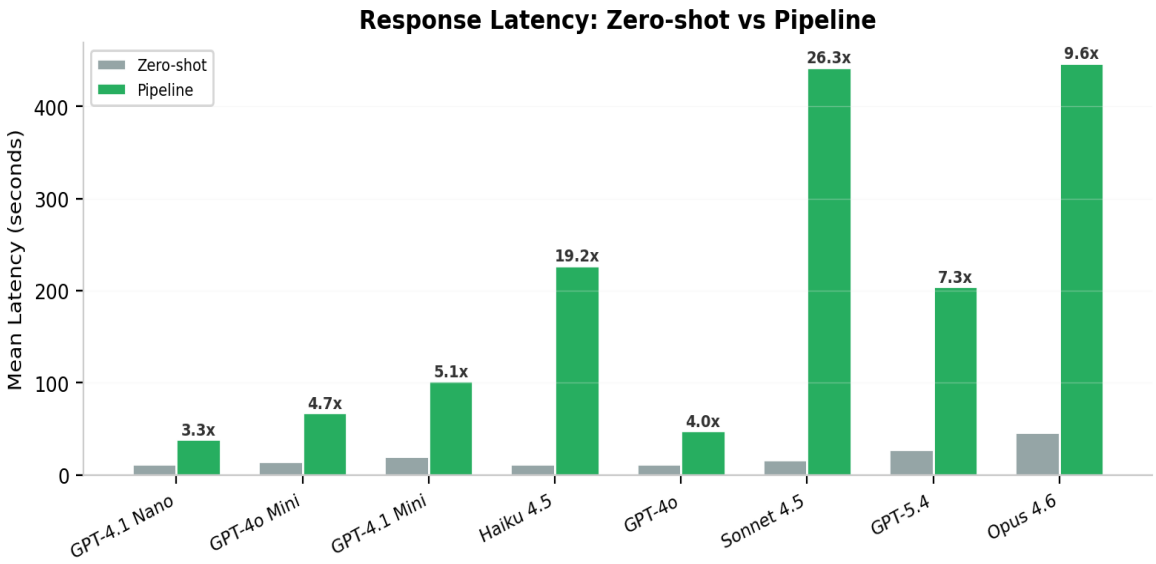

Figure 7. Response latency for zero-shot vs pipeline conditions. Labels show multiplier.

GPT-4.1 Nano pipeline completes in a mean of 39 seconds (3.3x its zero-shot latency). GPT-4.1 Mini takes 102 seconds (5.1x). Anthropic models are substantially slower: Haiku pipeline takes 227 seconds, and Opus pipeline takes 447 seconds. For synchronous user-facing applications, these latencies may be unacceptable. For asynchronous batch processing, they are less problematic.

5. Discussion

5.1 Why Scaffolding Helps Small Models

The data is consistent with the hypothesis that small models fail at complex decision tasks not because they lack the knowledge to reason well, but because they lack the discipline to deploy it systematically. They skip steps. They anchor on the first plausible answer. They do not self-critique. DODAR compensates by externalising the decision process.

This interpretation should be treated as a hypothesis supported by the data pattern, not as a proven mechanism. We did not measure what happens inside the models. The observation that scaffolding benefit correlates inversely with model capability is consistent with multiple causal stories: the models may genuinely lack reasoning discipline, or they may lack instruction-following capacity to execute the framework, or the framework may interact differently with different training distributions. The data cannot distinguish between these explanations.

5.2 Why the Pipeline Outperforms the Single Prompt

The pipeline amplifies the scaffolding effect because each phase receives dedicated model attention. In the single-prompt condition, the model must hold all five phases in working memory simultaneously. For small models, this creates competition between phases for limited context. The pipeline eliminates this by running each phase as a focused, independent call with the accumulated output from prior phases as input.

The specialised personas also appear to contribute. The pipeline does not simply split the prompt into five calls; it assigns each phase a role-specific system prompt that primes the model for a particular mode of thinking. Whether the benefit comes from the decomposition, the personas, or both cannot be determined from this evaluation. An ablation study stripping personas while keeping the decomposition would clarify this.

5.3 The Aviation Analogy

The analogy to aviation training is suggestive rather than rigorous. A new first officer has the technical knowledge to fly the aircraft but may lack the metacognitive discipline to manage a non-normal situation under pressure. CRM training, including DODAR, provides the external scaffold that prevents premature closure and forces systematic evaluation. Experienced captains internalise this discipline over time and no longer need the explicit framework.

A similar pattern appears in the data: frontier models appear to have internalised the reasoning discipline that small models need externally provided. This analogy is illustrative. It should not be taken as evidence of a shared mechanism between human cognition and language model computation.

6. Practical Applications

6.1 Production Deployment

The results suggest a deployment pattern for high-volume decision-support workloads: use a small model with a DODAR pipeline where latency tolerance permits. GPT-4.1 Mini pipeline offers the strongest quality-to-cost ratio, achieving 104% of frontier quality at 89% lower cost with a mean latency of 102 seconds.

For latency-sensitive applications, GPT-4.1 Nano pipeline completes in 39 seconds at 88% of frontier quality. For applications requiring maximum quality with no latency constraint, frontier models with zero-shot prompting remain the simpler and faster option.

6.2 Framework Design Principles

Several design principles emerge from the per-dimension analysis. First, include an explicit review phase. This is where the largest gains appear. Second, use specialised personas for each phase when implementing as a pipeline. Third, be cautious with the decision phase: the pipeline may fragment decision reasoning, so consider whether the decision justification can reference prior phases explicitly. Fourth, monitor for ceiling effects: if scores are consistently at or near maximum, the framework may be producing compliance rather than genuine reasoning improvement.

6.3 When Not to Use the Pipeline

The pipeline adds cost, latency, and architectural complexity. It is not appropriate for all use cases. For simple queries, routine tasks, or situations where a frontier model is already performing well, the pipeline adds overhead without proportionate benefit. The Sonnet anomaly demonstrates that near-frontier models may perform better with a verbose prompt than with imposed structure. The pipeline is most valuable when cost is a binding constraint, the task requires structured multi-step reasoning, and latency is tolerable.

7. Limitations

This section catalogues the methodological weaknesses of the evaluation. They are presented in approximate order of severity.

Ceiling effects and scale sensitivity. The 5-point scale cannot discriminate well at the top end. With 10.0% of cells scoring 4.90 or above, the scale is saturated for pipeline and frontier conditions. A wider scale would provide better resolution.

No human evaluation baseline. All scoring was automated. We have no human expert scores on these scenarios to anchor what a 4.80 on this scale means in practice. It is possible that both evaluators systematically overrate or underrate quality relative to human judgement.

Evaluator-model overlap. Despite dual scoring, the overlap problem persists. Claude evaluates Anthropic models. GPT-5.4 evaluates OpenAI models. Averaging partially mitigates but does not eliminate potential family-level bias.

Prompt sensitivity untested. A single version of each prompt was tested (v2 for most conditions). We do not know how sensitive results are to prompt wording. The GPT-5.4 collapse could be an artefact of one particular prompt formulation rather than a stable model property. Robustness testing across prompt variants is needed.

Small effective sample size. Each model-condition combination has only 10 underlying response items (one per scenario). The reported n=20 reflects dual-evaluator scoring, not 20 independent responses. The DODAR single-prompt n=40 reflects four scoring passes over the same 10 items. Apparent precision in the means may overstate confidence given the true independence is n=10.

Model selection bias. Only OpenAI and Anthropic models were tested. Results may not generalise to models from Google (Gemini), Meta (Llama), Mistral, or DeepSeek. These models have different training approaches and may respond differently to structured scaffolding.

GPT-5.4 as an unreliable data point. The extreme variance in GPT-5.4's scores under multiple conditions (std up to 1.83) means this model's data points are unreliable as benchmarks. Including it in frontier averages drags the baseline down. Excluding it changes the headline numbers.

Limited scenario count and unvalidated design. Only ten scenarios were fully evaluated, from a bank of twenty. The evaluated set was not randomly sampled. Scenarios were authored for this study and have not been independently validated for difficulty calibration, domain coverage, or discriminative power. Ten scenarios is a thin basis for generalisation.

Single framework tested. Only DODAR was evaluated. Other structured frameworks (OODA, Six Thinking Hats, pre-mortem analysis) might produce different scaffolding effects. The results should not be generalised to all reasoning frameworks.

Pipeline latency not optimised. The pipeline runs phases sequentially. Parallelisation of independent phases, prompt caching, or other optimisations could reduce latency substantially but were not tested.

Effect sizes based on single-prompt DODAR. The exported effect sizes (Cohen's d) compare single-prompt DODAR against baselines. Pipeline-specific effect sizes were not calculated, which understates the pipeline's statistical significance.

8. Conclusion

Structured reasoning frameworks, when applied as engineering scaffolds rather than pedagogical tools, function as cost-efficiency multipliers that enable small language models to approach frontier-level performance on complex decision tasks.

The DODAR pipeline is the strongest finding. GPT-4.1 Mini with the pipeline scores 4.80 out of 5.0, reaching 104% of Opus 4.6 zero-shot quality at 89% lower real API cost. The pipeline also reduces output variance, making results more consistent and predictable.

These findings come with substantial caveats. Ceiling effects may mask true performance differences. The evaluation lacks a human baseline. Prompt sensitivity is untested. The model sample is limited to two providers. The GPT-5.4 collapse may be a prompt artefact rather than a stable finding. These limitations do not invalidate the central finding, but they constrain how far it can be generalised.

For practitioners, the actionable insight is that model selection and prompt engineering interact strongly. The optimal production configuration is not always the most capable model. It is often the smallest model that can reliably follow the framework, paired with a pipeline that provides the reasoning discipline the model would not deploy on its own.

The framework originates from aviation, where structured decision-making under uncertainty is a safety-critical discipline. The same structural principles transfer to language models because the failure modes are analogous: premature closure, option narrowing, and absent self-critique. Structure compensates for capability gaps in both domains.

Appendix A: Scenario List

The scenario bank defines 20 scenarios. The 10 marked with an asterisk (*) were fully evaluated across all models and conditions and form the basis of all reported results. The remaining 10 received either partial or no evaluation.

ID	Title	Domain	Status
AMB-01	SaaS Churn Spike	Business	Fully evaluated *
AMB-02	Chest Pain in Acute & Emergency	Medical	Fully evaluated *
AMB-03	Clinical Negligence Enquiry — Causation Gap	Legal	Fully evaluated *
AMB-04	Intermittent 500 Errors in Production	Tech	Fully evaluated *
AMB-05	Exhaustion Despite 8 Hours Sleep	Personal	Fully evaluated *
AMB-06	Sudden App User Retention Cliff	Tech	Not evaluated
AMB-07	Unexplained Revenue Variance in Recurring Contract	Business	Not evaluated
AMB-08	Persistent Headache with Atypical Trigger	Medical	Not evaluated
AMB-09	Sudden Churn in Professional Services Contracts	Business	Not evaluated
AMB-10	Inconsistent Test Results Across Batches	Tech	Not evaluated
TRD-01	Pre-Seed Startup — Hire, Market, or Cut Burn	Startup	Fully evaluated *
TRD-02	Senior Engineer Considering CTO Role at Startup	Career	Fully evaluated *
TRD-03	City Council Green Belt Housing Decision	Policy	Fully evaluated *
TRD-04	Angel Investor Choosing Between 3 Opportunities	Investment	Fully evaluated *
TRD-05	Law Firm Build vs. Buy vs. Outsource for Docume...	Operations	Fully evaluated *
TRD-06	Acquired Founder Choosing Company Offer vs. Sta...	Career	Not evaluated
TRD-07	Hospital Administrator Choosing Orthopedic Unit...	Healthcare	Not evaluated
TRD-08	Nonprofit Board Deciding on Growth Strategy	Strategy	Not evaluated
TRD-09	Engineering Manager Choosing Between Two Candid...	Management	Not evaluated
TRD-10	Family Deciding Between Three School Options	Personal	Not evaluated

Table A1. Scenario bank. Only the 10 asterisked scenarios contributed to reported results.

Appendix B: Per-Model Detailed Scores

Tables B1 through B8 show the full score breakdown by model, condition, and dimension.

Table B1. GPT-4.1 Nano (Nano)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	3.30 (0.46)	3.15 (0.57)	4.28 (0.45)	4.15 (0.57)	3.85 (0.57)
Option Breadth	2.90 (0.62)	2.70 (0.56)	4.03 (0.57)	3.75 (0.54)	3.55 (0.50)
Decision Justif.	3.15 (0.65)	3.05 (0.67)	3.55 (0.71)	3.50 (0.59)	3.60 (0.58)
Action Specificity	2.65 (0.48)	2.70 (0.56)	3.75 (0.43)	4.50 (0.59)	3.20 (0.68)
Review / Self-Corr.	2.20 (0.40)	2.10 (0.70)	4.45 (0.55)	4.80 (0.40)	3.35 (0.57)
Overall Trust.	3.15 (0.48)	3.05 (0.59)	3.77 (0.69)	3.75 (0.43)	3.70 (0.56)
Average	2.89	2.79	3.97	4.08	**3.54**

Table B2. GPT-4o Mini (Small)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	3.00 (0.55)	3.05 (0.38)	3.85 (0.36)	3.85 (0.36)	3.25 (0.43)
Option Breadth	2.85 (0.65)	2.25 (0.43)	3.42 (0.49)	3.50 (0.50)	3.00 (0.45)
Decision Justif.	2.80 (0.60)	2.70 (0.46)	3.02 (0.61)	2.95 (0.50)	3.00 (0.71)
Action Specificity	2.80 (0.40)	2.20 (0.60)	3.42 (0.63)	3.90 (0.30)	2.50 (0.50)
Review / Self-Corr.	2.05 (0.50)	1.80 (0.68)	4.00 (0.45)	4.45 (0.50)	2.90 (0.54)
Overall Trust.	3.05 (0.50)	2.80 (0.51)	3.17 (0.54)	3.35 (0.48)	3.15 (0.36)
Average	2.76	2.47	3.48	3.67	**2.97**

Table B3. GPT-4.1 Mini (Small)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	3.75 (0.43)	3.75 (0.43)	4.85 (0.36)	5.00 (0.00)	4.55 (0.50)
Option Breadth	3.25 (0.70)	3.15 (0.73)	4.62 (0.48)	4.75 (0.43)	4.40 (0.73)
Decision Justif.	3.60 (0.58)	3.70 (0.64)	4.40 (0.49)	4.40 (0.73)	4.15 (0.65)
Action Specificity	3.20 (0.40)	3.10 (0.54)	4.45 (0.55)	4.95 (0.22)	3.90 (0.30)
Review / Self-Corr.	2.65 (0.57)	2.75 (0.43)	5.00 (0.00)	5.00 (0.00)	4.25 (0.54)
Overall Trust.	3.65 (0.48)	3.60 (0.49)	4.47 (0.50)	4.70 (0.46)	4.45 (0.59)
Average	3.35	3.34	4.63	4.80	**4.28**

Table B4. Haiku 4.5 (Small)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	4.00 (0.71)	4.35 (0.48)	4.95 (0.22)	4.90 (0.30)	4.75 (0.43)
Option Breadth	3.35 (0.79)	3.10 (0.62)	4.80 (0.40)	4.85 (0.36)	4.30 (0.71)
Decision Justif.	4.15 (0.73)	4.20 (0.51)	4.10 (0.58)	4.00 (0.45)	4.65 (0.57)
Action Specificity	4.05 (0.67)	3.75 (0.62)	3.67 (0.52)	4.60 (0.49)	4.75 (0.43)
Review / Self-Corr.	3.40 (0.97)	3.30 (0.71)	4.55 (0.50)	5.00 (0.00)	4.70 (0.46)
Overall Trust.	3.95 (0.74)	4.10 (0.44)	4.40 (0.49)	4.35 (0.48)	4.50 (0.59)
Average	3.82	3.80	4.41	4.62	**4.61**

Table B5. GPT-4o (Mid)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	3.55 (0.67)	3.75 (0.83)	4.20 (0.40)	4.20 (0.40)	3.70 (0.56)
Option Breadth	3.25 (1.04)	3.55 (1.12)	4.08 (0.65)	4.10 (0.54)	3.45 (1.07)
Decision Justif.	3.75 (0.94)	3.80 (0.93)	3.77 (0.76)	3.40 (0.66)	3.85 (0.79)
Action Specificity	3.15 (0.96)	3.15 (0.96)	3.67 (0.79)	4.15 (0.48)	3.25 (0.83)
Review / Self-Corr.	3.00 (1.10)	3.00 (1.10)	4.28 (0.45)	4.85 (0.36)	3.30 (0.78)
Overall Trust.	3.80 (0.87)	3.95 (0.86)	3.88 (0.71)	3.85 (0.48)	3.95 (0.74)
Average	3.42	3.53	3.98	4.09	**3.58**

Table B6. Sonnet 4.5 (Mid-Frontier)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	4.15 (0.91)	4.28 (0.67)	4.92 (0.26)	5.00 (0.00)	4.95 (0.22)
Option Breadth	3.90 (1.09)	3.65 (1.04)	4.65 (0.48)	5.00 (0.00)	4.80 (0.40)
Decision Justif.	4.55 (0.67)	4.42 (0.54)	4.83 (0.38)	4.45 (0.67)	4.95 (0.22)
Action Specificity	4.45 (0.67)	4.00 (0.97)	4.17 (0.54)	4.55 (0.59)	4.75 (0.43)
Review / Self-Corr.	4.10 (1.04)	3.65 (0.99)	4.55 (0.50)	5.00 (0.00)	4.85 (0.36)
Overall Trust.	4.40 (0.80)	4.30 (0.64)	4.72 (0.45)	4.50 (0.50)	4.90 (0.30)
Average	4.26	4.05	4.64	4.75	**4.87**

Table B7. GPT-5.4 (Frontier)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	4.50 (0.50)	4.55 (0.59)	1.80 (1.40)	5.00 (0.00)	3.00 (1.67)
Option Breadth	4.10 (0.83)	4.15 (1.06)	1.93 (1.62)	5.00 (0.00)	3.05 (1.77)
Decision Justif.	4.75 (0.43)	4.80 (0.40)	1.90 (1.58)	4.50 (0.50)	3.20 (1.83)
Action Specificity	4.40 (0.92)	4.30 (0.84)	1.75 (1.34)	4.55 (0.92)	2.80 (1.54)
Review / Self-Corr.	4.30 (0.90)	4.25 (0.89)	1.77 (1.35)	5.00 (0.00)	2.95 (1.66)
Overall Trust.	4.65 (0.48)	4.75 (0.43)	1.93 (1.62)	4.75 (0.43)	3.20 (1.83)
Average	4.45	4.47	1.85	4.80	**3.03**

Table B8. Opus 4.6 (Frontier)

Dimension	Zero-shot	CoT	DODAR	Pipeline	Length-m.
Diagnosis Quality	4.65 (0.48)	4.75 (0.43)	4.95 (0.22)	5.00 (0.00)	4.95 (0.22)
Option Breadth	4.30 (0.78)	4.05 (0.89)	4.95 (0.22)	5.00 (0.00)	4.55 (0.59)
Decision Justif.	4.75 (0.43)	4.88 (0.33)	4.85 (0.36)	4.65 (0.48)	4.95 (0.22)
Action Specificity	4.80 (0.40)	4.55 (0.59)	3.88 (0.68)	4.80 (0.40)	4.75 (0.54)
Review / Self-Corr.	4.45 (0.74)	4.28 (0.71)	4.50 (0.50)	5.00 (0.00)	4.65 (0.48)
Overall Trust.	4.75 (0.43)	4.65 (0.48)	4.85 (0.36)	4.75 (0.43)	4.85 (0.36)
Average	4.62	4.53	4.66	4.87	**4.78**

Appendix C: Pipeline Architecture

The DODAR pipeline decomposes the single-prompt instruction into five sequential model calls. Each call receives a phase-specific system prompt that includes a role persona, the original scenario, and the accumulated output from all prior phases. The architecture is as follows.

Phase 1: Diagnose. System prompt assigns a "diagnostic reasoning specialist" persona. Input is the raw scenario only. The model is instructed to generate competing hypotheses, surface assumptions, and identify missing information. Output is validated against a minimum hypothesis count before proceeding.

Phase 2: Options. System prompt assigns a "strategic options analyst" persona. Input is the scenario plus the Diagnose output. The model generates distinct alternatives with explicit trade-offs. Validated against a minimum option count.

Phase 3: Decide. System prompt assigns a "decision architect" persona. Input is the scenario plus Diagnose and Options outputs. The model commits to a recommendation with transparent justification against specific alternatives.

Phase 4: Action. System prompt assigns an "implementation planning specialist" persona. Input includes all prior phase outputs. The model produces concrete, sequenced implementation steps with dependencies, timelines, and reversibility assessments.

Phase 5: Review. System prompt assigns a "critical review analyst" persona. Input includes all prior phase outputs. The model identifies failure modes, validates assumptions from earlier phases, and specifies conditions for plan abandonment.

The context window grows with each phase. By Phase 5, the input includes the scenario text plus the full output of Phases 1 through 4. This context accumulation is the primary driver of the token multiplier, which ranges from 13x (GPT-4.1 Nano) to 49x (Haiku 4.5) relative to a single zero-shot call.

Code and data: github.com/afieldofdreams/dodar

Correspondence: adam@crox.io